Predicting H1N1 Vaccine Uptake Using Machine Learning

Project Overview

This project focuses on understanding and predicting **H1N1 vaccination behavior** using machine learning techniques.

The objective is to identify key factors that influence whether individuals choose to receive the vaccine.

By analyzing demographic characteristics, health behaviors, and perceptions about vaccines, we aim to uncover patterns that can support **data-driven public health decision-making**.

The model developed in this project is designed to help stakeholders better understand vaccination trends and improve the effectiveness of targeted health interventions.

Business Understanding

- Despite the availability of vaccines, a significant portion of the population does not get vaccinated due to factors such as lack of awareness, low perceived risk, or mistrust in vaccine effectiveness.
- Public health organizations often struggle to identify which individuals or groups are less likely to vaccinate, making it difficult to design efficient and targeted vaccination campaigns.
- This leads to **inefficient use of resources** and increases the risk of disease spread.

Key Question:

How can we predict vaccination behavior to better target public health interventions?

Stakeholders:

- Kenya Ministry of Health — responsible for national vaccination strategies
- Africa Centres for Disease Control and Prevention — supports disease prevention across Africa
- World Health Organization — guides global vaccination policies

Data Understanding

The dataset used in this analysis contains survey responses from **26,707 individuals**, capturing a wide range of factors that may influence vaccination behavior.

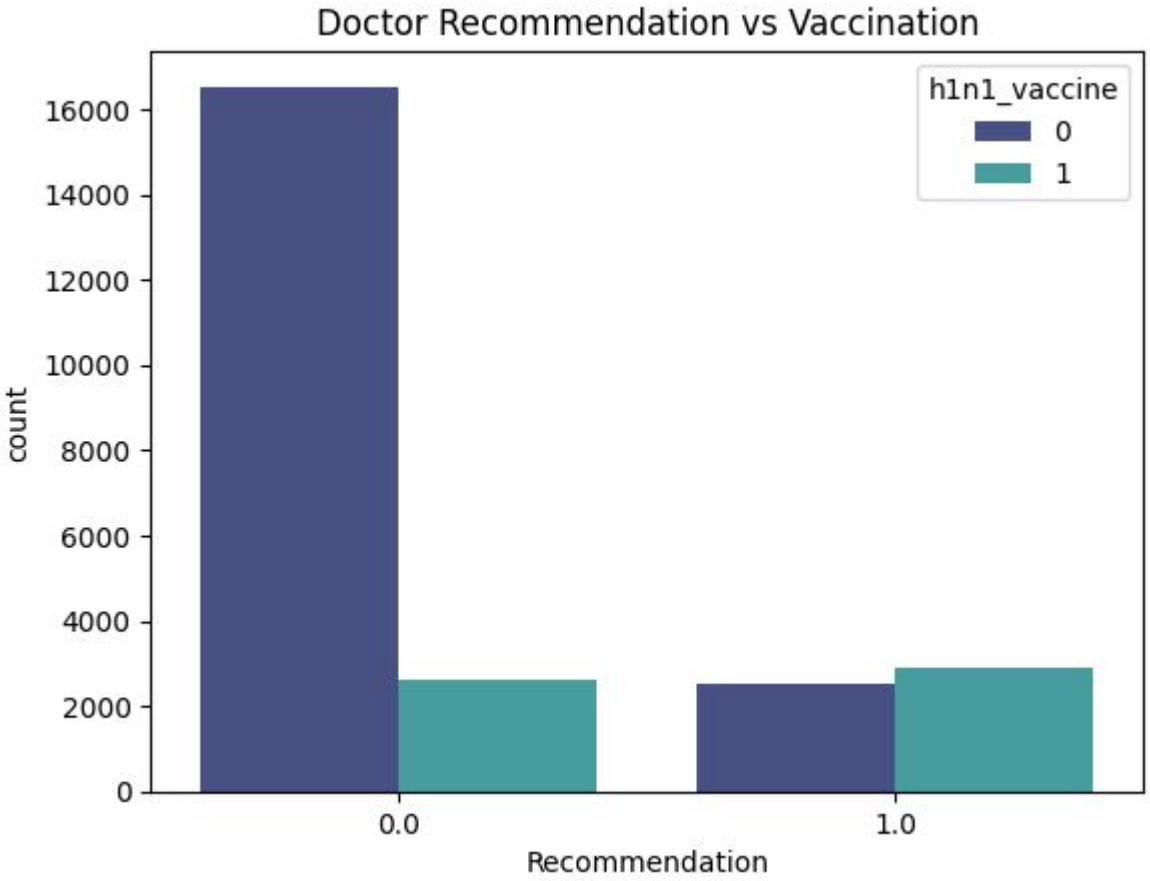
These factors include:

- demographic characteristics such as age, education, and income
- health-related behaviors such as preventive practices
- personal opinions and perceptions about the H1N1 virus and vaccines

The target variable represents whether an individual received the H1N1 vaccine, making this a **binary classification problem**.

Understanding the structure and composition of this dataset is essential for identifying meaningful patterns and preparing the data for modeling.

Key Insights - EDA



The chart displays that **individuals who received a doctor's recommendation are significantly more likely to be vaccinated, indicating that medical advice plays a crucial role in vaccine uptake**.

Other insights :

- Belief in vaccine effectiveness increases uptake
- Higher concern → higher vaccination
- Health workers vaccinate more

Modelling

To address the business problem, a classification modeling approach was used to predict vaccination outcomes.

A **Logistic Regression model** was selected as the baseline due to its interpretability and suitability for binary classification problems.

An improved version of the model was developed by addressing class imbalance and tuning key parameters to enhance performance.

The models were evaluated using metrics such as **recall, precision, and ROC-AUC,** with particular emphasis on recall to ensure that vaccinated individuals are accurately identified.

This iterative approach allowed for continuous improvement and better alignment with the project objectives

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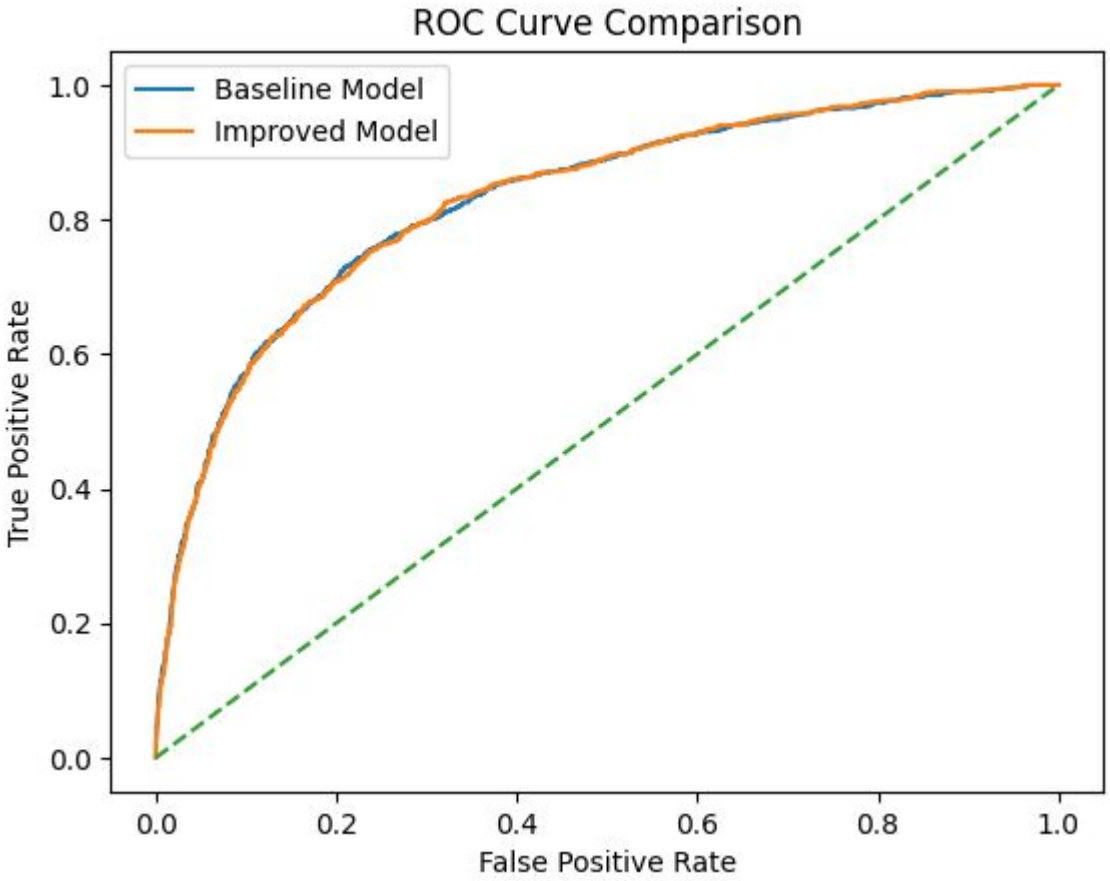
Model Performance



- Both models demonstrated similar overall predictive capability, but **the improved model** is more aligned with the project goal of understanding vaccination behavior.
- The improved model correctly identifies approximately 72% of vaccinated individuals, compared to only 44% in the baseline model.

Key Drivers of Vaccination

Doctor recommendation emerged as one of the most significant drivers, emphasizing the importance of healthcare engagement.

Individuals who believe the vaccine is effective are more likely to receive it, highlighting the **role of public trust and perception.**

Personal concern about the H1N1 virus also plays a major role, as individuals who perceive higher risk are more likely to take preventive action.

Occupation, particularly being a health worker, further influences vaccination likelihood due to increased exposure and awareness.

These factors collectively demonstrate that vaccination decisions are influenced by both **information and perception.**

Recommendations

Strengthen communication between healthcare providers and patients by encouraging doctors to actively recommend vaccination.

Develop public health campaigns that emphasize the safety and effectiveness of vaccines to improve public confidence.

Increase awareness of the risks associated with H1N1 to influence perception and encourage preventive behavior.

Target outreach efforts toward individuals outside the healthcare sector who may have less access to reliable medical information.

Tailor vaccination campaigns to specific demographic groups to improve engagement and effectiveness.

These actions can help improve vaccination rates and support better public health outcomes.

Future Work

Incorporate more recent and region-specific data, particularly from African populations, to improve the relevance of the model.

Explore more advanced machine learning techniques to further enhance predictive performance.

Integrate the model into public health systems to support real-time decision-making and targeted interventions.

Continuously monitor and update the model to adapt to changing public health conditions and behaviors.

Thank You

Thank You for your attention.

Feel free to reach out if you have any questions to :

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